

Edeviser — Tech Stack, AI & Product Cost Overview

Quick Reference for Founders, Team Members & Investors

Version: 1.0 | **Date:** April 2026

What Is Edeviser?

Edeviser is a B2B SaaS platform for higher education that fuses accreditation compliance (OBE) with student engagement (gamification) through a Dual-Engine Architecture. When a teacher grades an assignment, the system simultaneously generates accreditation evidence and rewards the student with XP, badges, and streak progress. Compliance and engagement feed each other in a closed loop.

Target market: Private and semi-government universities in Qatar and the GCC region, with expansion into South/Southeast Asia. The platform supports full Arabic/English bilingual operation with RTL layout.

Complete Tech Stack

Frontend

Technology	Role
React 18 + TypeScript (strict mode)	UI framework
Vite 6	Build tooling, HMR, bundling
Tailwind CSS v4 + Shadcn/ui	Styling and component library
TanStack Query v5	Server state management, caching
TanStack Table	Data tables with sort/filter/pagination
React Router v7	Role-based routing and guards
React Hook Form + Zod	Forms and runtime validation
Zustand	Client-side state management
Framer Motion	Animations and micro-interactions
Recharts	Charts and data visualization
i18next + react-i18next	Internationalization (Arabic/English)
Sentry	Error monitoring and performance tracing
Sonner	Toast notifications
nuqs	URL-persisted search/filter state
dnd-kit	Drag and drop (rubric builder, outcome reordering)
date-fns	Date manipulation
canvas-confetti	Celebration effects (level-up, badge unlock)

Backend (Supabase BaaS)

Technology	Role
PostgreSQL 15	Primary database with Row Level Security on all tables
Supabase Auth (GoTrue)	JWT-based authentication, session management
Supabase Realtime	WebSocket-based live updates (leaderboard, grades, notifications)
Supabase Storage	File uploads (assignments, avatars) with RLS
Supabase Edge Functions (Deno)	Server-side business logic (25+ functions)
pgvector	Vector embeddings for AI RAG pipeline
pg_cron	Scheduled jobs (streak reset, at-risk signals, weekly summaries)
pg_net	HTTP calls from database triggers
PostgREST	Auto-generated REST API from database schema

External Services

Service	Role	Monthly Cost (Pilot)
Vercel Pro	Frontend CDN, CI/CD, preview deployments	\$20
OpenRouter (GPT-4o)	AI features (at-risk, suggestions, feedback, quizzes)	\$60–80
OpenAI Embeddings	RAG vector embeddings (text-embedding-3-small/large)	\$2–5
Resend Pro	Transactional email (streak reminders, grade notifications)	\$20
Sentry	Production error monitoring	\$29

Dev Tooling

Tool	Role	Monthly Cost
Claude Code (Anthropic Max)	AI-assisted development	\$200
Kiro IDE Max	Spec-driven development	\$39
GitHub Enterprise Cloud (2 users)	Source control, CI/CD, Copilot	\$42
CodeRabbit Pro	Automated PR review	\$19
Figma Professional (2 editors)	UI/UX design and handoff	\$30
Postman Basic	API testing	\$14

Testing

Tool	Role
Vitest	Unit and integration test runner
fast-check	Property-based testing (min 100 iterations)
Testing Library	React component testing
Playwright	End-to-end testing
k6	Load testing

AI Features & Insights

Edeviser runs 6 AI-powered features, all production-ready and backed by a feedback flywheel that improves accuracy over time.

1. At-Risk Early Warning

- Analyzes behavioral signals (login frequency, submission timing, CLO attainment trends) from the student activity log
- Flags students with high probability of failing a CLO, 7+ days before deadlines
- Each prediction includes a probability score (0–100%) and contributing signal details
- Surfaced on the teacher dashboard with actionable nudge options
- Validated against actual grades for accuracy tracking (target: <20% false positive rate)

2. Personalized Module Suggestions

- Identifies CLO attainment gaps per student and recommends specific learning resources
- Includes social proof from historical cohort data (e.g., "Students who completed these exercises scored 34% higher on this CLO")
- Student feedback (thumbs up/down) stored and fed back into model weighting

3. Assignment Feedback Drafts

- Generates per-criterion rubric feedback comments based on submitted work
- Teacher sees draft comments with accept/edit/reject controls per criterion
- Available within 10 seconds of opening the grading view
- Teacher actions stored for model improvement

4. Adaptive Quiz Generation

- AI generates quiz questions aligned to specific CLOs and Bloom's Taxonomy levels
- Question bank with difficulty calibration, discrimination index tracking, and quality flags
- Adaptive quiz sessions adjust difficulty in real-time based on student performance
- Post-quiz review with AI-generated explanations and confidence scoring

5. AI Tutor (RAG Pipeline) — Designed

- Retrieval-Augmented Generation using pgvector for course-specific knowledge
- Students ask questions and receive contextual answers grounded in course materials
- Designed and spec'd; deployment planned for pilot phase

6. Habit Correlation Analysis

- Computes correlations between student habits (login, submission, journaling, reading) and academic outcomes
- Surfaces insights like "Students who journal 3+ times per week score 22% higher on CLO attainment"
- Confidence badges on correlation strength

AI Feedback Flywheel

Every AI output collects structured feedback. Student ratings, teacher accept/edit/reject actions, and actual grade outcomes are stored in the `ai_feedback` table. This data feeds back into prompt engineering and model selection, creating a continuous improvement loop.

AI Architecture Decisions

- LLM calls routed through OpenRouter for provider flexibility (GPT-4o primary, fallback to Claude)
 - All AI processing happens in Supabase Edge Functions (server-side, never client-side)
 - Rate limiting on all AI endpoints to control costs
 - AI features are modular — each can be enabled/disabled per institution
-

Product Cost Summary

Monthly Infrastructure (Pilot: 1 institution, 500 students)

Category	Monthly (USD)	Monthly (PKR)
Supabase Pro + compute	\$50	PKR 14,000
Vercel Pro	\$20	PKR 5,600
AI/LLM APIs (GPT-4o)	\$60–80	PKR 16,800–22,400
AI Embeddings	\$2–5	PKR 560–1,400
Resend Pro (email)	\$20	PKR 5,600
Sentry (monitoring)	\$29	PKR 8,120
Infrastructure total	~\$181–206	~PKR 50,680–57,680

Monthly Dev Tooling

Category	Monthly (USD)	Monthly (PKR)
Claude Code + Kiro + GitHub + CodeRabbit + Figma + Postman	\$344	PKR 96,320

A single developer with this AI-augmented tooling stack maintains velocity equivalent to a 3-person team. Each tool replaces the need for additional engineering hires.

Combined Monthly Burn

Scenario	Total/month (PKR)	Total/month (USD)
Lean (essential tools only)	~PKR 130,000	~\$464
Standard (all tools)	~PKR 151,000	~\$539
Growth (3 institutions, 2K users)	~PKR 175,000	~\$625

Scaling Path

Scale	Users	Infrastructure/mo
Pilot	500	~\$180/mo
Early Growth	2,000	~\$280/mo
Growth	10,000	~\$535/mo
Enterprise (50K+)	50,000+	Migration to AWS/self-hosted Supabase

Capital Requirement

PKR 2M bridge capital covers Qatar market entry (PKR 1.4M for founder relocation, QFC entity, regulatory compliance) plus 6 months of product infrastructure (PKR 624K). The Qatar setup cost is reimbursable through the Ignite Bridge Startup Program at month 6-7, extending the effective runway to 12+ months with a PKR 606K buffer.

Revenue Model

Tier	Students	Annual License	Implementation Fee
Starter	Up to 500	\$30K–50K	\$10K
Growth	500–2,000	\$50K–100K	\$20K
Enterprise	2,000+	\$100K–250K	\$40K

A single discounted pilot contract (~PKR 4.2M/year) exceeds the entire PKR 2M bridge capital and makes the company self-sustaining on infrastructure costs.

Why Supabase (BaaS) Instead of AWS / Custom Node.js Backend?

This is the most common technical question from investors and engineers. Here is the honest comparison.

The Short Answer

Supabase gives a 2-person team the backend capabilities of a 6-person team at 1/10th the cost. For a pre-revenue startup targeting first pilot, speed to market matters more than architectural purity. The migration path to AWS exists and is well-defined if we outgrow it.

Option A: Supabase BaaS (What We Chose)

Pros

- Auth, database, realtime, storage, edge functions, and cron jobs in a single managed service — no infrastructure to maintain
- Row Level Security (RLS) enforces data isolation at the database layer, not in application code. This is stronger security than most custom backends achieve
- PostgREST auto-generates a typed REST API from the database schema — zero boilerplate CRUD code
- Realtime subscriptions (WebSocket) built in — leaderboards, grade notifications, and streak updates work out of the box
- pgvector extension for AI embeddings, pg_cron for scheduled jobs, pg_net for HTTP triggers — all native PostgreSQL, no extra services
- Edge Functions run on Deno (TypeScript) — same language as the frontend, no context switching
- Open-source core — can self-host on AWS/GCP if needed, eliminating vendor lock-in
- \$50/month for a production-grade backend that would cost \$500-1,000/month on AWS with equivalent services
- Built-in dashboard for database management, user auth, storage, and logs — no need for separate admin tools
- Type generation from schema (`supabase gen types`) keeps frontend and database in sync automatically

Cons

- Edge Functions have cold start latency (~200-500ms on first invocation) — not ideal for latency-critical endpoints
- No custom middleware layer — complex request pipelines require workarounds in Edge Functions
- Limited to PostgreSQL — no easy path to NoSQL or multi-database architectures
- Supabase Realtime has connection limits (~200 concurrent per project on Pro plan) — needs upgrade at scale
- Less granular control over database tuning, connection pooling, and query optimization compared to self-managed PostgreSQL
- Community and ecosystem smaller than AWS/Firebase — fewer third-party integrations
- Edge Function debugging is harder than a local Node.js server — limited logging and no step-through debugging

Option B: AWS (Lambda + RDS + API Gateway + Cognito)

Pros

- Virtually unlimited scale — handles millions of concurrent users without architectural changes

- Fine-grained control over every infrastructure component (networking, caching, queuing, compute)
- Mature ecosystem with hundreds of managed services (SQS, SNS, ElastiCache, CloudFront, etc.)
- Enterprise compliance certifications (SOC 2, HIPAA, FedRAMP) built in
- Multi-region deployment for global low-latency access
- Extensive monitoring and observability (CloudWatch, X-Ray, CloudTrail)

Cons

- Requires a dedicated DevOps engineer or significant infrastructure expertise — a 2-person team cannot maintain this
- Monthly cost for equivalent functionality: \$500-1,500/month minimum (RDS, Lambda, API Gateway, Cognito, S3, CloudFront, SQS)
- Auth (Cognito) is notoriously difficult to configure and has poor developer experience compared to Supabase Auth
- No built-in realtime — requires separate WebSocket infrastructure (API Gateway WebSocket + Lambda + DynamoDB)
- No built-in RLS equivalent — data isolation must be implemented in application code, which is error-prone
- Cold starts on Lambda (especially with VPC) can be 1-3 seconds
- Development velocity is 3-5x slower due to infrastructure configuration, IAM policies, and deployment pipelines
- Vendor lock-in is deeper — migrating away from Lambda + DynamoDB + Cognito is harder than migrating away from Supabase (which is just PostgreSQL)

Option C: Custom Node.js/Express Backend

Pros

- Full control over every line of backend code — no platform constraints
- Familiar to most developers — largest hiring pool
- Can use any database, any ORM, any middleware
- No cold starts — always-on server with predictable latency
- Easier debugging with standard Node.js tooling

Cons

- Must build and maintain: auth system, session management, RBAC, file upload handling, realtime WebSocket server, cron job scheduler, database migrations, API documentation, rate limiting, input validation — all from scratch
- Estimated 3-4 months of backend-only development before the first feature ships
- Requires server management (or Kubernetes/ECS) — ongoing DevOps burden
- Security is only as good as the code — no database-layer enforcement like RLS
- Monthly hosting: \$100-300/month for a basic setup (VPS/container + managed DB + Redis + S3)
- A 2-person team would spend 60-70% of time on backend infrastructure instead of product features

The Migration Plan (When We Outgrow Supabase)

Trigger	Action	Timeline
50+ institutions or 50K+ users	Migrate to self-hosted Supabase on AWS (same codebase, own infrastructure)	2-3 weeks
Need for custom middleware/queuing	Add API Gateway + SQS in front of Supabase for specific endpoints	1-2 weeks
Enterprise compliance requirement	Move to AWS GovCloud with self-hosted Supabase	4-6 weeks
Realtime connection limits hit	Add dedicated WebSocket server (Socket.io on ECS) alongside Supabase	1-2 weeks

The key insight: Supabase is PostgreSQL. Every query, every migration, every RLS policy works identically on self-hosted PostgreSQL. The migration is an infrastructure change, not a code rewrite.

Cost Comparison (Monthly, for Edeviser at Pilot Scale)

Component	Supabase	AWS Equivalent	Custom Node.js
Database	Included (\$50 plan)	RDS: \$50-100	Managed DB: \$30-60
Auth	Included	Cognito: \$0-25	Build from scratch
Realtime	Included	API GW WebSocket + Lambda: \$30-50	Socket.io server: \$20-40
Storage	Included	S3 + CloudFront: \$10-20	S3: \$5-10
Edge Functions	Included	Lambda + API GW: \$20-50	VPS/Container: \$20-50
Cron Jobs	Included (pg_cron)	EventBridge + Lambda: \$5-10	node-cron on server
DevOps time	0 hours/month	20-40 hours/month	10-20 hours/month
Total	\$50/month	\$150-300/month + DevOps	\$100-200/month + DevOps

Bottom Line

For a pre-revenue startup with a 2-person team, Supabase is the right choice today. It lets us ship a production-grade platform at \$50/month with enterprise-level security (RLS), built-in realtime, and zero DevOps overhead. The migration path to AWS is clean because Supabase is just PostgreSQL underneath. We revisit this decision at 50+ institutions — and by then, we will have the revenue and team size to justify the infrastructure investment.

Architecture Highlights (Investor-Relevant)

- **Security at the database layer:** Row Level Security on every table. No application-layer trust. Data isolation is enforced by PostgreSQL, not by code.
- **Immutable audit trail:** Evidence records and XP transactions are append-only. Accreditation auditors can trace any attainment score back to the original submission.
- **Real-time everything:** Leaderboards, grade notifications, streak updates, and habit tracking all update live via WebSocket subscriptions.
- **Multi-tenancy via RLS:** Single database, institution-scoped data isolation. Scales to 50+ institutions before needing architectural changes.
- **Open-source backend:** Supabase is open-source. Migration path to self-hosted or AWS exists if needed, eliminating vendor lock-in risk.
- **PWA-ready:** Service worker, offline queue, and manifest.json for installable mobile experience without app store overhead.

Key Numbers

Metric	Value
Feature modules	17+
Supabase Edge Functions	25+
Database tables with RLS	30+
Database migrations	40+
React hooks	120+
Shared UI components	50+
Test files (unit + property)	150+
Supported languages	Arabic, English (full RTL)
User roles	5 (admin, coordinator, teacher, student, parent)
AI features	6 (5 live, 1 designed)
XP sources	11+
Badge types	Standard, mystery, tiered, team
Cron jobs	10 (streak, at-risk, summaries, habits, etc.)

Competitive Moat

1. **Dual-engine fusion** — No competitor combines OBE compliance and gamification in a single feedback loop
2. **AI-native from day one** — At-risk prediction, adaptive quizzes, and feedback drafts are core features, not add-ons
3. **Science-backed design** — 10 foundational pillars grounded in published research (Bloom, Fogg, Zimmerman, Csikszentmihalyi, Eyal, Chou)
4. **Qatar-first localization** — Full Arabic/English bilingual with RTL layout, targeting a \$7.2B GCC EdTech market with no dominant OBE platform
5. **Low infrastructure cost** — Full production platform runs at ~\$180/mo, enabling aggressive pricing against enterprise competitors